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Project Synopsis Template (2025-26) - Sem VII

JeevanSetu

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Abstract

Non-Governmental Organizations (NGOs) working in the domain of child welfare play a vital role in addressing the social, emotional, and educational needs of underprivileged and homeless children in India. However, many of these organizations struggle with outdated operational systems, lack of automation, and minimal digital infrastructure, which collectively hinder their ability to function efficiently, scale operations, and build long-term trust with stakeholders. Key areas such as donation tracking, volunteer management, campaign execution, and logistics coordination are often managed manually or through disjointed tools, leading to inefficiencies, data mismanagement, and transparency issues.

This project proposes a comprehensive, full-stack web-based platform designed to transform the way child support NGOs operate by providing an integrated system that addresses their core operational challenges. The platform offers secure and transparent donation management through Razorpay payment gateway integration and blockchain-based transaction tracking, enabling both financial transparency and donor confidence. It includes robust modules for volunteer coordination, such as QR code-based attendance, incentive tracking, and campaign management tools, which streamline outreach and engagement. Additionally, dedicated features for logistics management—including food distribution, housing arrangements, and medical supply tracking—ensure the efficient handling of on-ground operations.

One of the key features of the system is a dynamic analytics dashboard that provides insights into all aspects of NGO activities, from donations and drives to volunteer participation and impact metrics. This enhances decision-making and reporting, while also strengthening accountability to external stakeholders such as donors, government bodies, and the community.

By adopting this platform, NGOs can significantly enhance their operational efficiency, build stronger relationships with donors and volunteers, and most importantly, increase their ability to positively impact the lives of vulnerable children. This project serves as a step toward digitizing social impact and bringing technological empowerment to the nonprofit sector.

Introduction

Non-Governmental Organizations (NGOs) serve as the backbone of civil society, especially in the field of child welfare, where they provide essential services such as shelter, education, health care, and emotional support to vulnerable children. In India, the demand for such services is significant due to the large number of homeless and underprivileged children who lack access to basic needs. NGOs like Jeevan Samvardhan are committed to rehabilitating these children and helping them become self-reliant members of society.

Despite their critical role, many child-centric NGOs face major hurdles in scaling their operations efficiently. These challenges are often rooted in outdated or non-integrated systems for managing donations, coordinating volunteers, organizing drives, and handling daily logistics. Manual processes lead to data inconsistencies, limited transparency, and difficulty in reporting outcomes to donors and governing bodies.

In an age where digital transformation is redefining the functioning of sectors across the globe, NGOs too must adapt and adopt technology-driven approaches to improve their reach and impact. A centralized, digital management system can address these pain points by automating repetitive tasks, offering real-time insights through dashboards, and creating secure and auditable systems for financial and operational workflows.

Problem Statement

Child-focused NGOs in India play a crucial role in supporting underprivileged and homeless children, yet many face significant operational challenges due to outdated and fragmented systems.

Donation management is a major pain point, with many NGOs lacking secure, centralized platforms to process and track contributions in real time. This leads to delayed fund access, limited financial accountability, and reduced donor confidence. Volunteer coordination also suffers due to the absence of systematic registration, attendance tracking, and engagement tools, causing communication gaps and lowered participation in drives. Logistics and resource distribution—covering essentials like food, clothing, shelter, and medical supplies—are frequently unmonitored in real time, resulting in wastage, shortages, and poor allocation during critical events.

NGOs struggle with the lack of integrated analytics and reporting systems to evaluate performance, forecast needs, and present impact reports to governing bodies or donors. The absence of technology-driven solutions also limits their ability to deliver structured educational content through volunteers or implement advanced initiatives like missing child identification using facial recognition. Without a centralized, secure, and intelligent digital platform, these challenges collectively restrict the scalability, transparency, and overall social impact of child support NGOs.

“ This project seeks to build such a platform an all-in-one solution that integrates donation management, volunteer coordination, drive logistics, and analytics into a seamless interface. By leveraging technologies like blockchain, Razorpay integration, and web-based dashboards, this system aims to not only streamline operations but also foster accountability and encourage more active participation from donors and volunteers alike. ”

Proposed Solution

JeevanSetu aims to address the operational challenges faced by child support NGOs, this project proposes the development of a comprehensive full-stack web-based platform that integrates donation management, volunteer coordination, logistics handling, and analytics into a unified system. The objective is to streamline workflows, improve transparency, and enable data-driven decision-making by leveraging modern and reliable technologies.

1. Secure Donation Management with Transparency

One of the primary features of the proposed system is a secure and transparent donation management module. The platform will facilitate online donations through integration with Razorpay, a widely used payment gateway in India. This will enable NGOs to accept payments via UPI, debit/credit cards, net banking, and other digital payment methods directly into their bank accounts.

To further enhance the security and credibility of the system, blockchain technology will be utilized to ensure that each transaction is recorded in a decentralized, immutable ledger. This implementation will make all donation records tamper-proof and easily auditable. Donors will have access to a clear transaction history and will be able to track how their contributions are being utilized, fostering trust and long-term engagement.

2. Volunteer and Drive Management System

In addition to managing donations, the platform will streamline the planning and execution of NGO-led **drives and campaigns** such as food distribution, medical outreach, and educational programs. NGO administrators will be able to:

- **Create and publish drives or events**, detailing time, location, and resource requirements.
- Allow volunteers to **register and participate** in events through the platform.
- Track volunteer **attendance using QR codes**, ensuring accurate and efficient record-keeping.
- **Incentivize volunteers** through recognition programs or point-based systems to encourage continued engagement.

This module will reduce the administrative burden of coordinating logistics and manpower while improving volunteer accountability and engagement.

3. Logistics Tracking and Resource Management

Effective logistics are essential for the success of any field operation. The platform will offer tools to manage and monitor key resources such as **housing, food, clothing, and medical supplies**. NGOs will be able to:

- Track inventory levels and usage across different campaigns.
- Monitor the distribution of supplies in real time.
- Forecast future resource requirements based on past data and upcoming drives.

This functionality will support efficient allocation of resources and minimize waste, ensuring that aid reaches those who need it most.

4. Comprehensive Analytics and Reporting Dashboard

To provide visibility into operations and support data-driven decision-making, the platform will include a **real-time analytics dashboard**. This dashboard will compile data from all core modules and present it in a user-friendly visual format. Key features will include:

- Customizable reports based on defined timeframes (daily, weekly, monthly).
- Insights into donation patterns and funding sources.
- Volunteer participation metrics and performance indicators.
- Campaign performance tracking and comparative analysis.
- Inventory and resource usage reports.

Such insights will assist NGO leadership in evaluating program effectiveness, optimizing operations, and reporting outcomes to donors, partners, and regulatory bodies.

5. Educational Modules for Volunteer-Led Teaching

To support NGOs in their efforts to provide informal education to children, the platform will optionally include a learning module tailored for volunteers who wish to participate as educators. This feature is designed to enable structured and consistent delivery of basic educational content by non-professional teaching volunteers.

Key functionalities of this module will include:

- Access to curated teaching materials covering foundational subjects such as literacy, numeracy, health awareness, and life skills.
- Age-appropriate and multilingual content designed specifically for underprivileged children.
- Guided lesson plans and activity kits to assist volunteers with little or no prior teaching experience.
- Optional training content for volunteers, helping them understand child psychology, effective communication, and best practices for teaching in community settings.

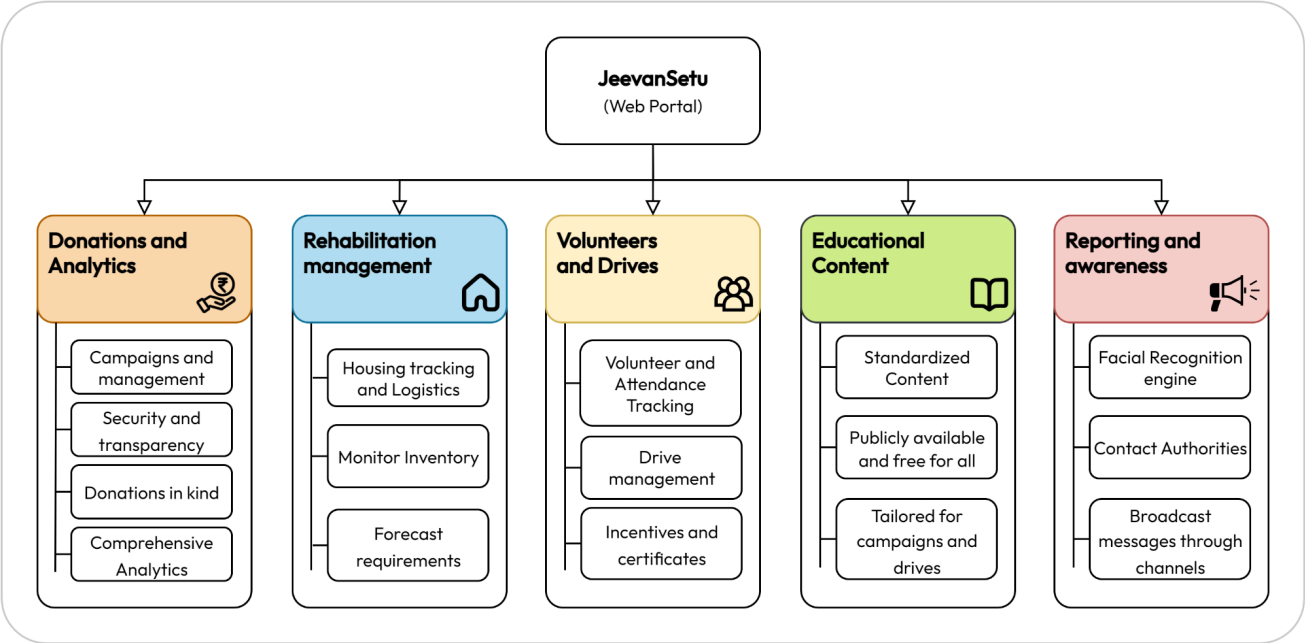
6. Missing Child Search and Reporting

To assist NGOs working with vulnerable and displaced children, the platform will include a dedicated module for **missing child identification and reporting**. This feature will leverage **facial recognition technology** to help match reported missing children with existing records or uploaded images, improving the chances of identification and reunification.

Key components include:

- A **facial recognition engine** that compares uploaded photos with a database of recovered or reported children to find potential matches.
- A secure and searchable **missing child registry**, allowing NGOs to report new cases and update statuses.
- Option for the public or other NGOs to **submit tips or reports** with supporting information or photos.

Block Diagram and Explanation



The block diagram outlines five core modules of the proposed NGO management platform. It includes **Donations and Analytics** for secure fundraising and insights, **Rehabilitation Management** for handling housing and resources, and **Volunteers and Drives** to manage campaigns and track volunteer participation. The **Educational Content** module provides accessible teaching materials for outreach, while **Reporting and Awareness** supports missing child identification, authority alerts, and public messaging. Together, these modules streamline NGO operations and enhance social impact.

Hardware, Software, and Tools Requirements

Hardware Requirements:

- **Developer Machine:**
  - Processor: Intel i5 (10th Gen or newer) / AMD Ryzen 5
  - RAM: 16 GB
  - Storage: 512 GB SSD or more

- **User Machine:**
  - Processor: Intel i3 (10th Gen or newer)
  - RAM: 8 GB
  - Storage: 256 GB SSD or more

Software Requirements:

- **Frontend:**
  - React (v18.2.0)
  - Material UI (v5.14.8)

- **Backend:**
  - Node.js (v20.5.1)
  - Express (v4.18.2)
  - Flask (v2.0.1)
- **Database:**
  - MongoDB (v8.0)
- **Payment Gateway:**
  - Razorpay API
- **Blockchain Layer:**
  - Ethereum Testnet / Hyperledger (for transaction immutability)

Tools Used:

- Design Tools: Figma, Canva
- Version Control: Git, GitHub
- Project Management: Trello / Jira

Proposed Evaluation Measures

To measure the effectiveness of the proposed system, we will use the following evaluation criteria:

1. **Donation Tracking Accuracy**
    - Ensure every donation made through Razorpay is logged accurately in both the backend and blockchain.
    - Perform audits to compare platform records with bank statements to ensure consistency.

2. **Volunteer Engagement Metrics**
    - Track the number of volunteers joining drives, total attendance via QR scan, and repeat participation.
    - Use incentive tracking (points, badges) to evaluate motivation and contribution levels.

3. **System Usability and Performance**
    - Measure frontend and backend response times.
    - Monitor server uptime, load testing under high usage, and mobile/browser compatibility.

4. **Security and Data Integrity**
    - Blockchain implementation will ensure immutability of donation records.
    - Conduct penetration testing and vulnerability scans to ensure platform security.

5. **Logistics and Drive Management Efficiency**
    - Track the number of successful drives and resource distribution events managed using the platform.
    - Monitor reduction in errors or missed resources due to the new system.

6. **Analytics Usage and Decision Impact**
    - Evaluate how often NGO admins access analytics dashboards.
    - Collect feedback to understand whether analytics are helping in making better planning or funding decisions.

7. **Stakeholder Satisfaction**
    - Use feedback forms from donors, volunteers, and NGO staff to measure satisfaction.
    - Use Net Promoter Score (NPS) or simple rating scales to quantify user experience.

Conclusion

This project addresses the operational, financial, and logistical inefficiencies faced by child support NGOs through a single digital platform. By integrating modern technologies like blockchain, Razorpay, and full-stack web development frameworks, the system enhances trust, efficiency, and engagement among all stakeholders. This solution can significantly uplift the NGO ecosystem by digitizing and streamlining core operations.

References

[1] C. Nairi, M. Cicioğlu, and A. Çalhan, *"Smart Blockchain Networks: Revolutionizing Donation Tracking in the Web 3.0," Computer Communications*, vol. 2024, Art. no. 107972, 2024. doi: 10.1016/j.comcom.2024.107972. [Online]. Available: https://arxiv.org/abs/2311.03573

[2] S. Borade, P. Pagare, M. Shinde, and P. Suryawanshi, *"NGO and Donor Management System Using Charity Blockchain," Int. J. Innov. Res. Manag. Pharm. Sci. (IJIRMPS)*, vol. 12, no. 2, pp. 1–X, Mar.–Apr. 2024. [Online]. Available: http://www.ijirmps.org\

[3] B. Ata, M. H. Tongarlak, D. Lee, and J. Field, *"A Dynamic Model for Managing Volunteer Engagement,"* arXiv preprint arXiv:2312.02920v2 [stat.AP], Dec. 7, 2023. doi: 10.48550/arXiv.2312.02920

[4] P. Mikołajczak and P. Bajak, "Does NGOs' Commercialization Affect Volunteer Work? The Crowding Out or Crowding In Effect," Public Organization Review, 2020. doi: 10.1007/s11115-020-00478-1

[5] K. Shah, *"Introducing Simplified Online Donations & 80G Receipts for NGOs,"* Razorpay Blog, Aug. 21, 2020. [Online]. Available: https://razorpay.com/blog/accept-online-donations-with-80g-receipts/ Razorpay

[6] S. Saeed, M. Rohde, and V. Wulf, *"Designing IT Systems for NGOs: Issues and Directions,"* Dept. of Information Systems and New Media, University of Siegen, Germany. [Online]. Available:https://www.researchgate.net/publication/221002962_Designing_IT_Systems_for_NGOs_Issues_and_Directions

[7] Z. R. Shi, C. Wang, and F. Fang, *"Artificial Intelligence for Social Good: A Survey,"* arXiv preprint arXiv:2001.01818v1 [cs.CY], Jan. 7, 2020. doi: 10.48550/arXiv.2001.01818

[8] K. C. Dash and U. Mishra, "Critical Considerations for Developing MIS for NGOs," *arXiv preprint* arXiv:1403.7344 [cs.CY], Mar. 27, 2014. [Online]. Available: https://doi.org/10.48550/arXiv.1403.7344

[9] K. R. Varshney and A. Mojsilović, "Open Platforms for Artificial Intelligence for Social Good: Common Patterns as a Pathway to True Impact," *arXiv preprint* arXiv:1905.11519 [cs.CY], May 27, 2019. [Online]. Available: https://doi.org/10.48550/arXiv.1905.11519

[10] M. Bharadwaj, N. Prakash, and J. Thomas, "Web-Based Technology Interventions for Social Change: Insights from NGOs in Assam," *Journal of Propulsion Technology (Tuijin Jishu)*, vol. 44, no. 4, pp. 4165–4173, 2023.